Course Title: Technology, Environment and Society

HUM 4147

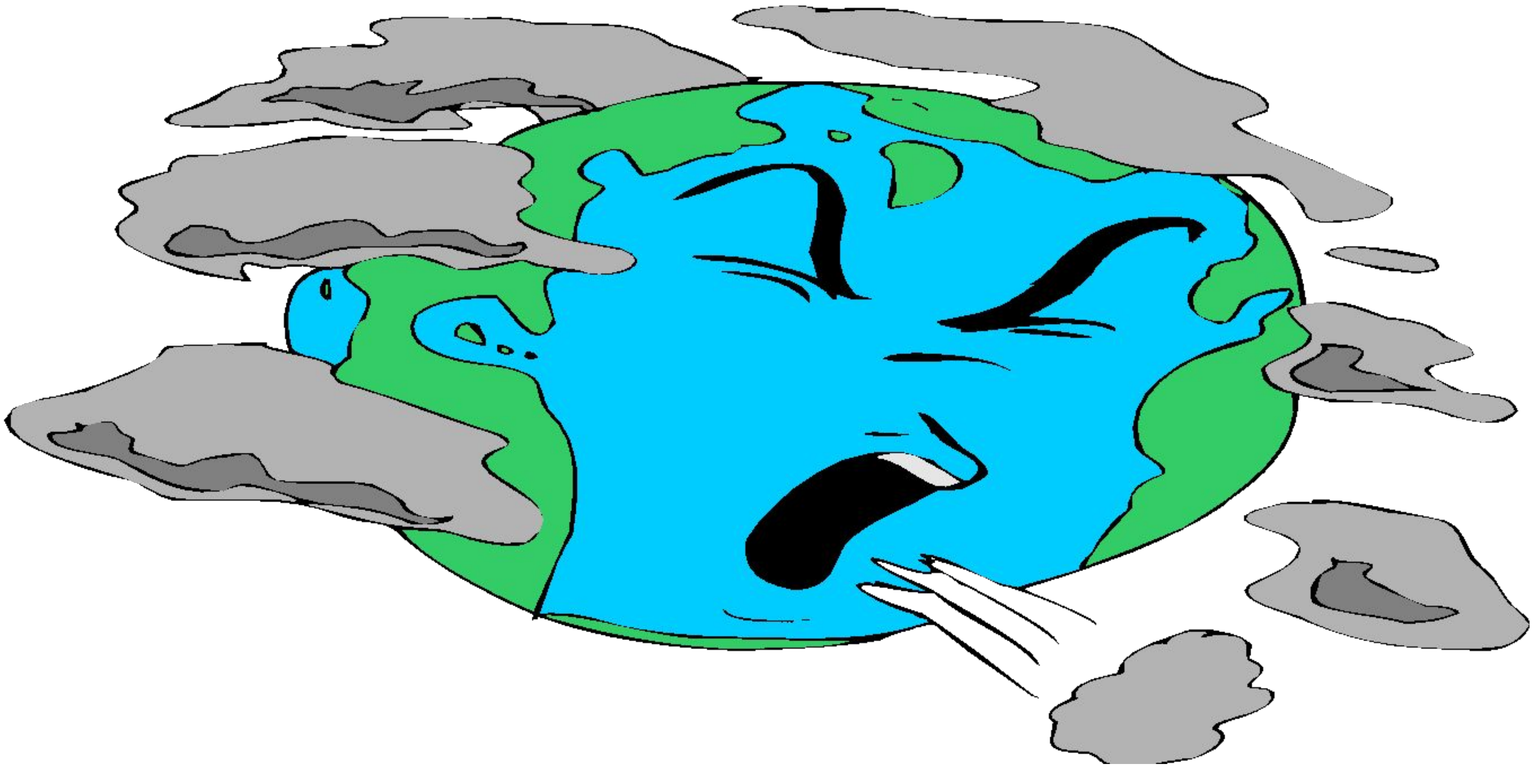
Content

Air Pollution

Air Pollution

Air Pollution

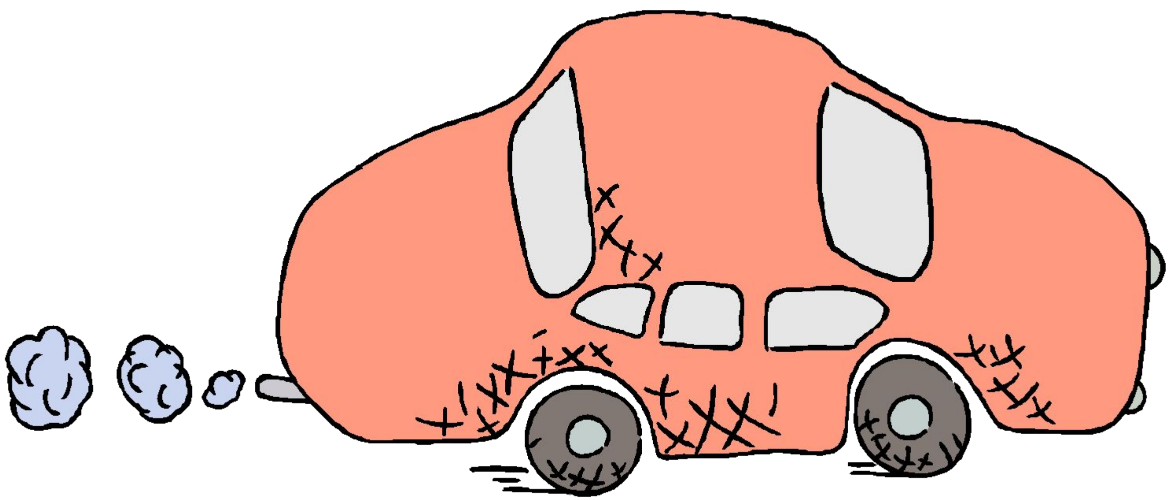
Any visible or invisible particle or gas found in the air that is not part of the original, normal composition.



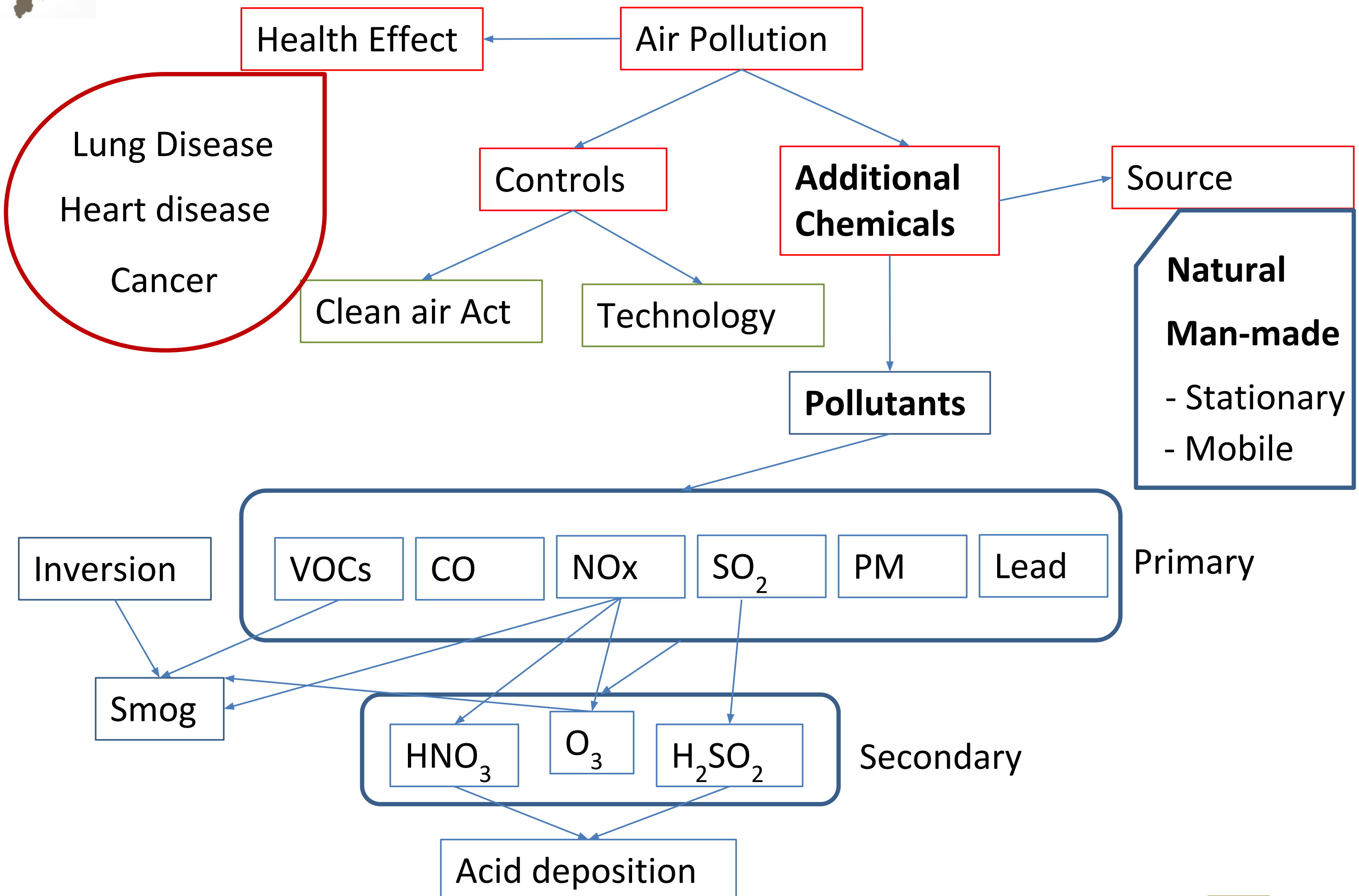
Air Pollution

Natural: forest fires, pollen, dust storm

Unnatural: man-made; coal, wood and other fuels used in cars, homes, and factories for energy



Air Pollution



Source: Bozeman Science

Air Pollution

5 Major Pollutants:

1.) Carbon Monoxide

2.) Sulfur Dioxide

3.) Nitrogen Dioxide

4.) Particulate Matter

5.) Ground Level Ozone

Air Pollution

Carbon Monoxide

•colorless, odorless gas

•produced when carbon does not burn in fossil fuels

•present in car exhaust

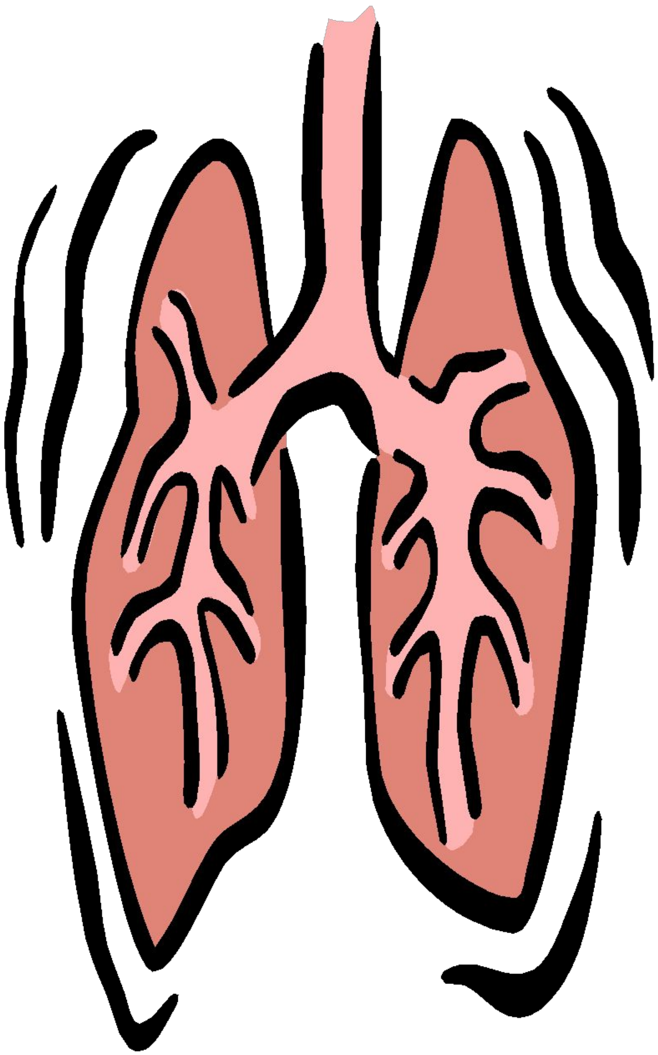
•deprives body of O2 causing headaches, fatigue, and impaired vision

Air Pollution

Sulfur Dioxide

- produced when coal and fuel oil are burned
- present in power plant exhaust
- narrows the airway, causing wheezing and shortness of breath, especially in those with asthma

Air Pollution



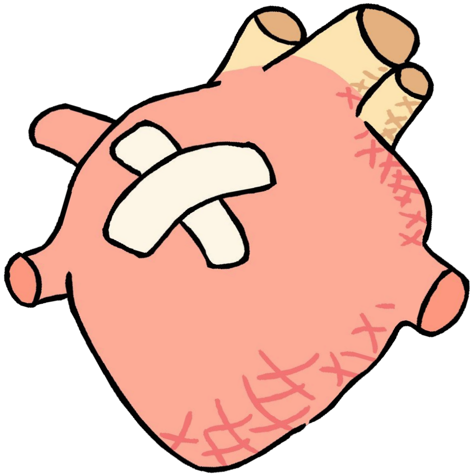
Nitrogen Dioxide

- reddish, brown gas
- produced when nitric oxide combines with oxygen in the atmosphere
- present in car exhaust and power plants
- affects lungs and causes wheezing; increases chance of respiratory infection

Air Pollution

Particulate Matter

- particles of different sizes and structures that are released into the atmosphere
- present in many sources including fossil fuels, dust, smoke, fog, etc.
- can build up in respiratory system
- aggravates heart and lung disease; increases risk of respiratory infection



Air Pollution

Ground Level Ozone

- at upper level, ozone shields Earth from sun's harmful UV rays
- at ground level, ozone is harmful pollutants
- formed from car, power and chemical plant exhaust
- irritate respiratory system and asthma; reduces lung function by inflaming and damaging lining of lungs

Air Pollution

SMOG

- Combination of gases with water vapor and dust
- Combination of smoke and fog
- Forms when heat and sunlight react (photochemical smog)
- Occurs often with heavy traffic, high temperatures, and calm winds

Air Pollution

Other Problems…

- ✔ Green House Effect
- ✔Global Warming
- ✔Thinning of Upper Ozone Layer
- ✔Acid Rain
- ✔Indoor Air Pollution

AQI: Air Quality Index

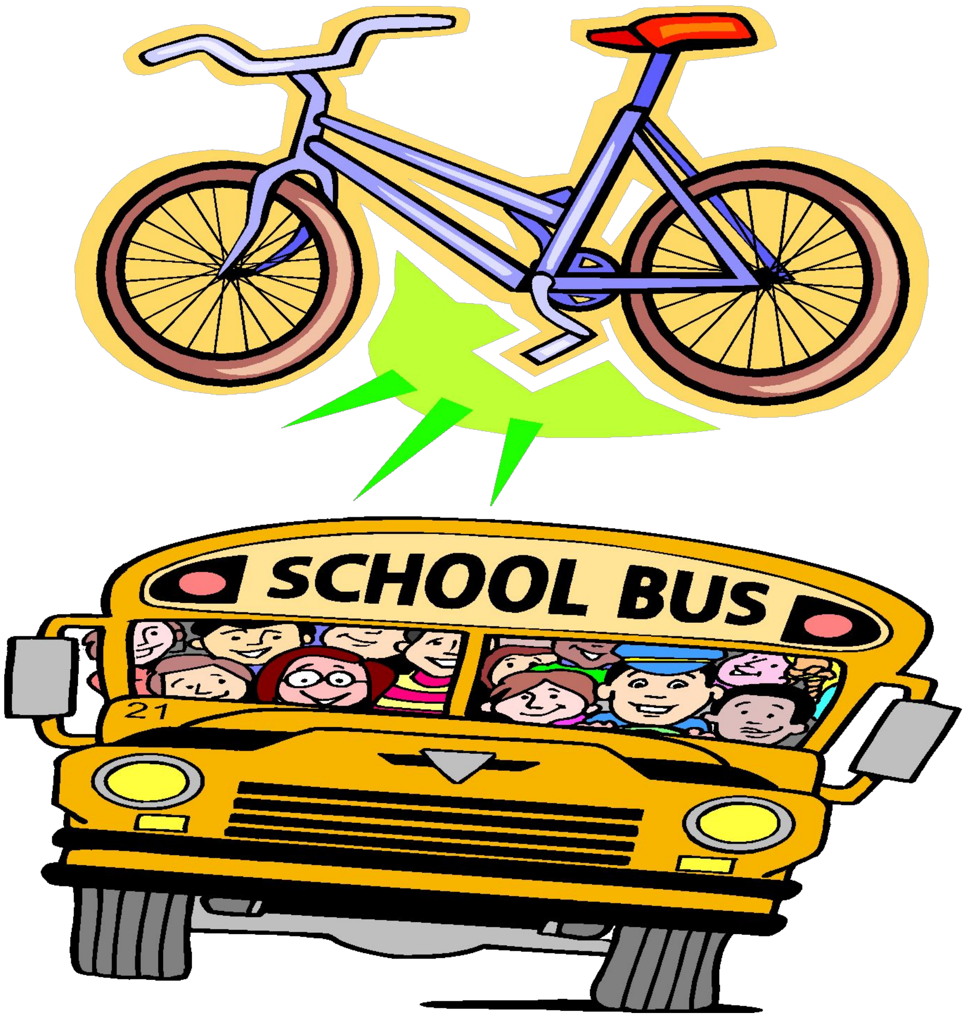
- The Air Quality Index (AQI) is **used for reporting daily air quality**.
- Indicates pollutant level in the air.
- It tells how clean or polluted air is, and what associated health effects might be a concern.
- The AQI focuses on health effects you may experience within a few hours or days after breathing polluted air.
- Ranges from 0 (least concern) to 500 (greatest concern)

Air Pollution

Air Quality Index Levels of Health Concern	Numerical Value	Meaning
Good	0-50	Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	51-100	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.
Unhealthy for Sensitive Groups	101-150	Members of sensitive groups may experience health effects. The general public is not likely to be affected.
Unhealthy	151-200	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects.
Very Unhealthy	201-300	Health alert: everyone may experience more serious health effects.
Hazardous	> 300	Health warnings of emergency conditions. The entire population is more likely to be affected.

What can we do?

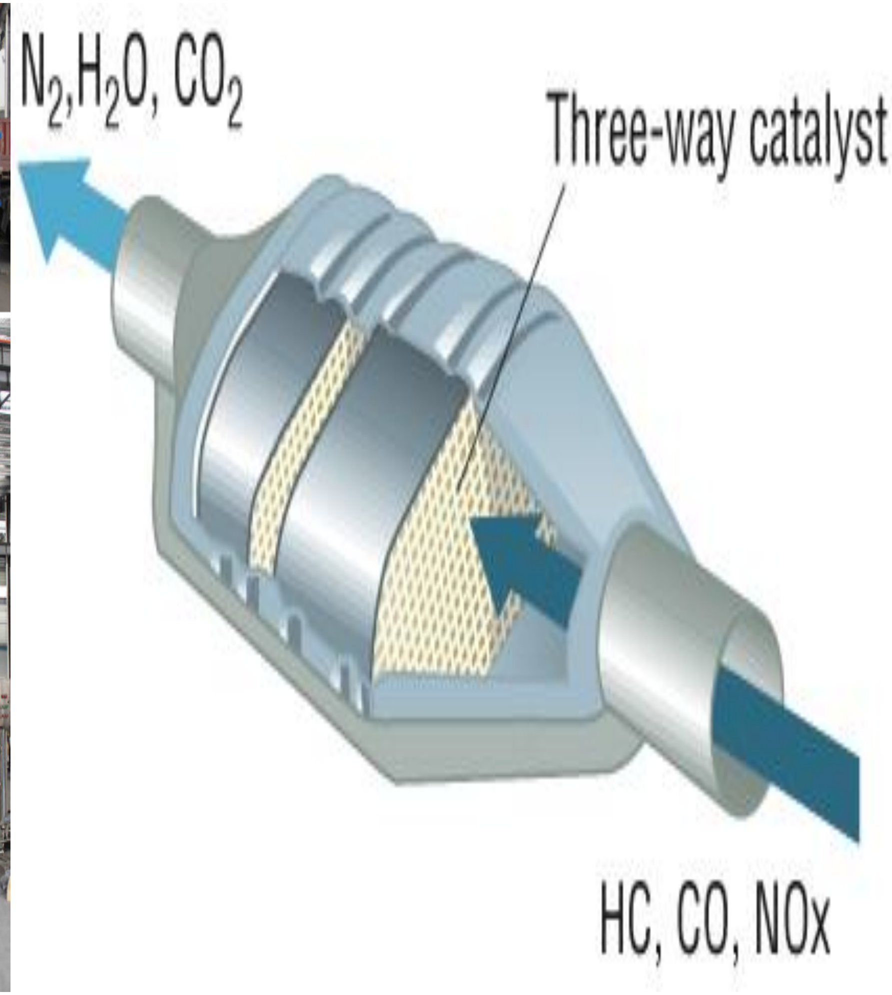
- Ride your bike
- Tell your friends and family about pollution
- Make sure your parents get pollution checks on their cars
- Ride the school bus



Air Pollution

- Learn more; stay up to date
- Join a group to stop pollution
- Encourage your parents to carpool to work
- Switch off lights, fan etc. when you leave the room/house.

Air Pollution



Mechanical filter

https://www.youtube.com/watch?v=6IJt2Z0pQDk

Catalytic Converter

Mechanical filter

Mechanical filter

A mechanical filter is a type of filter that is primarily designed to remove suspended solid particles. It can be used to maintain the cleanliness of a system, protecting it from abrasive wear or corrosion. Mechanical filters can be made of a fine mesh of synthetic polymer fibers.

A mechanical filter can be extremely useful in industrial settings such as in cooling systems where the environment can be extremely corrosive.

They are used for air-purifying respirators that mechanically stops particulates from reaching the wearer's nose and mouth. They come in multiple physical forms. With a properly functioning mechanical filter, industrial problems such as contamination and faults can be avoided, making mechanical filters an important part of industrial operations. With a clean and operational system, problems with corrosion can be prevented.

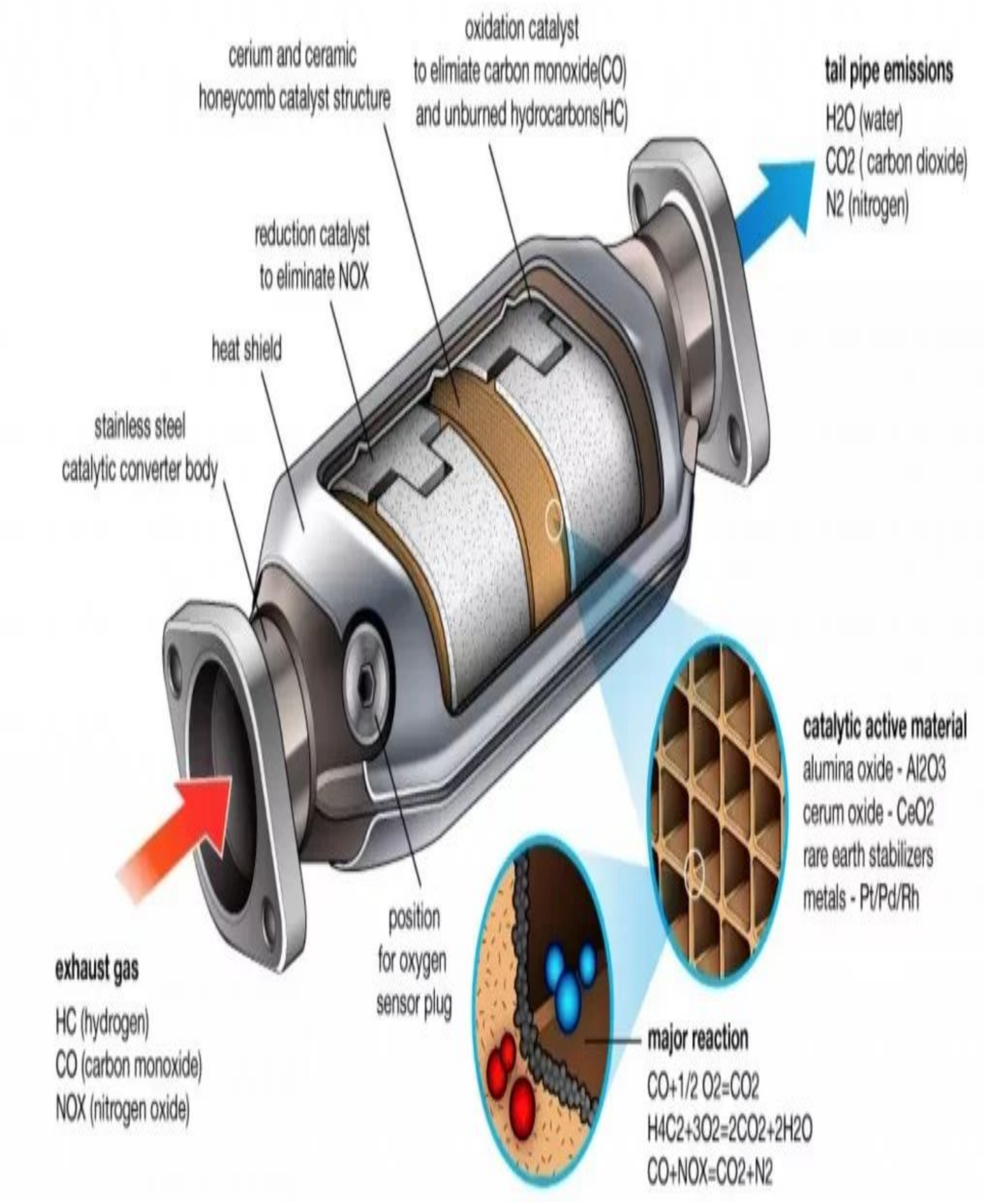
Mechanical filters remove contaminants from air in the following ways:-

1. by *interception* when particles following a line of flow in the airstream come within one radius of a fiber and adhere to it;
2. by *impaction*, when larger particles unable to follow the curving contours of the airstream are forced to embed in one of the fibers directly; this increases with diminishing fiber separation and higher air flow velocity
3. by an enhancing mechanism called *diffusion*, where gas molecules collide with the smallest particles, especially those below 100 nm in diameter, which are thereby impeded and delayed in their path through the filter; this effect is similar to Brownian motion and increases the probability that particles will be stopped by either of the two mechanisms above; it becomes dominant at lower air flow velocities.

4. by using electret filter material (usually, electrospun plastic fibers) to attract or repel particles with an electrostatic charge, so that they are more likely to collide with the filter surface
5. by using certain coatings on the fibers that kill or deactivate infectious particles colliding with them (such as salt)
6. by using gravity and allowing particles to settle into the filter material (this effect is typically negligible); and
7. by using the particles themselves, after the filter has been used, to act as a filter medium for other particles.

Catalytic converter

A catalytic converter is a unit that fits into the front part of exhaust system of an engine, **to reduce the emission of gaseous pollutants, such as carbon monoxide (CO), nitrogen oxides (NOx) and hydrocarbons (HCs)**.its an exhaust emission control device that converts toxic gases
and pollutants in exhaust
gas from an internal combustion engine into less-toxic pollutants by catalyzing a redox reaction.



Catalytic converters are usually used with internal combustion engines fueled by gasoline or diesel, including lean-burn engines, and sometimes on kerosene heaters and stoves. The catalyst formulation comprises three key constituents: precious metals (platinum, palladium, rhodium), alumina, and rare-earth-based materials which enhance catalytic activity of the metals. These "two-way" converters combine oxygen with carbon monoxide (CO) and unburned hydrocarbons (HC) to produce carbon dioxide (CO_2) and water (H_2O). Although catalytic converters are most commonly applied to exhaust systems in automobiles, they are also used on electrical generators, forklifts, mining equipment, trucks, buses, locomotives, motorcycles, and on ships. They are even used on some wood stoves to control emissions. This is usually in response to government regulation, either through environmental regulation or through health and safety regulations.